Solution:

(i) Variables: Let x_1 , x_2 and x_3 be the number of units of products P_1, P_2 and P_3 to be produced.

(ii) Objective function: Profit on x_1 units of the product $P_1 = 20\ x_1$

Profit on x_2 units of the product $P_2 = 25\ x_2$

Profit on x_3 units of the product $P_3 = 15\ x_3$

Total profit = $20\ x_1 + 25\ x_2 + 15\ x_3$

Since the total profit is to be maximized, we have to maximize $Z = 20\ x_1 + 25\ x_2 + 15\ x_3$

Constraints: $6x_1 + 3\ x_2 + 12\ x_3 \leq 200$

$2\ x_1 + 5\ x_2 + 4\ x_3 \leq 350$

$x_1 + 2\ x_2 + x_3 \leq 100$

Non-negative restrictions: Since the number of units of the products A,B and C cannot be negative, we have $x_1, x_2, x_3 \geq 0$

Thus, we have the following linear programming model.

Maximize $Z = 20\ x_1 + 25\ x_2 + 15\ x_3$

Subject to $6\ x_1 + 3\ x_2 + 12\ x_3 \leq 200$

$2x_1 + 5x_2 + 4\ x_3 \leq 350$

$x_1 + 2\ x_2 + x_3 \leq 100$

$x_1, x_2, x_3 \geq 0$

Example 10.3

A dietician wishes to mix two types of food F_1 and F_2 in such a way that the vitamin contents of the mixture contains atleast 6 units of vitamin A and 9 units of vitamin B. Food F_1 costs ₹ 50 per kg and F_2 costs ₹ 70 per kg. Food F_1 contains 4 units per kg of vitamin A and 6 units per kg of vitamin B while food F_2 contains 5 units per kg of vitamin A and 3 units per kg of vitamin B. Formulate the above problem as a linear programming problem to minimize the cost of mixture.

Solution:

(i) Variables:

Let the mixture contains x_1 kg of food F_1 and x_2 kg of food F_2

(ii) Objective function:

Cost of x_1 kg of food $F_1 = 50\ x_1$

Cost of x_2 kg of food $F_2 = 70x_2$

The cost is to be minimized

Therefore minimize $Z = 50\ x_1 + 70x_2$

(iii) Constraints:

We make the following table from the given data

Resources	Food (in kg)		Requirement
	F_1 (x_1)	F_2 (x_2)	
Vitamin A (units/kg)	4	5	6
Vitamin B (units/kg)	6	3	9
Cost (₹/kg)	50	70	

Table 10.1

$4\ x_1 + 5x_2 \geq 6$ (since the mixture contains 'atleast 6' units of vitamin A, we have the inequality of the type ≥)

$6\ x_1 + 3x_2 \geq 9$(since the mixture contains 'atleast 9' units of vitamin B ,we have the inequality of the type ≥)

(iv) Non-negative restrictions:

Since the number of kgs of vitamin A and vitamin B are non-negative, we have $x_1, x_2 \geq 0$

Thus, we have the following linear programming model

Minimize $Z = 50\ x_1 + 70x_2$ subject to

$4\ x_1 + 5x_2 \geq 6$

$6\ x_1 + 3x_2 \geq 9$ and $x_1, x_2 \geq 0$

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